



Semester: III
Subject Name: Data Analytics
Date: 09/16/2026 | Time: 1 Hour

JSS MAHAVIDYAPEETHA
JSS University, Noida
CIE-1 (AY 2026-27: Odd Semester)



Course: MCA
Subject Code: 25MCA320
Max. Marks: 20

Roll No.

COURSE OUTCOMES:

- CO1: Understand the fundamentals, sources and characteristics of data along with the need for data analytics.
CO2: Explain the key roles and phases involved in the Data Analytics Lifecycle.
CO3: Apply statistical techniques such as regression and Bayesian methods for data analysis.
CO4: Analyze advanced data modeling techniques such as neural networks, SVM and fuzzy logic.
CO5: Evaluate analytic results to support effective decision making.

PART A: ANSWER ALL THE QUESTIONS

Each question carries 1mark

4Qx1=04

Q	QUESTIONS	BL	CO
1	A Define Data Analytics.	1	CO1
	B What is semi-structured data? Give one example.	1	CO1
	C Name any two key roles required in a data analytics project team.	1	CO2
	D Define Principal Component Analysis (PCA).	2	CO4

PART B: ANSWER ALL THE QUESTIONS

Each question carries 4 marks

3QX4=12

Q	QUESTIONS	BL	CO
2	a Explain the evolution of analytic scalability, comparing traditional data processing platforms with modern Big Data platforms.	2	CO1
	b Distinguish between data analysis and reporting with suitable examples.	4	CO1
3	a Explain the Discovery and Data Preparation phases of the Data Analytics Lifecycle.	2	CO2
	b Numerical: For the following data on hours studied (X) and marks scored (Y): X: 1, 2, 3, 4, 5 Y: 2, 4, 5, 4, 5 Calculate the slope (b) and intercept (a) of the linear regression line $Y = a + bX$ using the least squares method, and predict Y for $X = 6$.	3	CO3
4	a Explain the working of a Support Vector Machine (SVM) for classification with a neat diagram.	2	CO4
	b Explain the concept of fuzzy logic and describe how fuzzy decision trees are extracted from data.	2	CO4

PART C: ANSWER ANY 1 QUESTION

Each question carries 4 marks

1Qx4=4

Q	QUESTIONS	BL	CO
5	Explain, with the help of a neat diagram, the six phases of the Data Analytics Lifecycle. Describe the key activities carried out in each phase.	4	CO2
6	Numerical: In a certain population, 1% of people have a particular disease. A diagnostic test for the disease gives a positive result 95% of the time for people who actually have the disease (true positive rate), and gives a positive result 10% of the time for people who do not have the disease (false positive rate). If a person selected at random tests positive, use Bayes' theorem to calculate the probability that the person actually has the disease.	5	CO3